ZAINAB YOUSAF

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PROFESSIONAL SUMMARY

- Full-Stack AI Engineer with 2+ years of experience building production AI agents, RAG pipelines and voice AI for customer communication across hospitality and e-commerce.
- Currently Full-Stack AI Engineer at DreamDesk AI (UK, remote), building voice, WhatsApp and chat agents live across 3 hotels.
- Research collaboration with DFKI Germany on smart contract security. Strong in agent architecture, MCP integrations and end-to-end delivery. Open to remote roles with UK and US overlap.

TECHNICAL SKILLS

AI & Agents: Pydantic AI, LangChain, LangGraph, RAG, MCP Servers, Tool Calling, Prompt Engineering, OpenAI, Gemini
Voice AI: VAPI, Retell, Text-to-Speech, Voice Cloning
Backend: Python, FastAPI, Node.js, NestJS, TypeScript, REST APIs, Webhooks
Data: PostgreSQL, MongoDB, Redis, Pinecone
Infra & Observability: Docker, GCP Cloud Run, GitHub Actions, Logfire, PostHog
Frontend & Mobile: Next.js, React, React Flow, Swift, SwiftUI

WORK EXPERIENCE

Full-Stack AI Engineer — Jan 2026 – Present

DreamDesk AI – Remote, UK (Full-time)

- Took ownership of the architecture and feature scoping; built multi-channel AI agents across voice calls, WhatsApp, email and web chat, live across 3 hotels.
- Built voice agents on **VAPI** and **Retell**, optimizing tool calls and MCP integrations to cut end-to-end response time to ∼1.6s, including knowledge-base retrieval and Mews PMS lookups that keep reservations in sync in real time; built the knowledge-base ingestion pipeline with conflict detection.
- Delivered the POC in 4 weeks, owning it end to end from requirements and feature design to testing; built analytics tracking AI cost per response, latency and agent performance for full operational visibility.
- **Tech Stack:** FastAPI, Next.js, Pydantic AI, Pinecone, PostgreSQL, VAPI, Retell, GCP Cloud Run, GitHub Actions

Software Engineer — Jul 2025 – Dec 2025

AutoMato AI – Lahore, Pakistan (Link)

- Built a GenAI pipeline that automated Amazon product listings for 50+ companies and sellers, cutting manual effort by 40–60%.
- Integrated Amazon Seller API to pull live competitor data and used OpenAI to extract high-ranking keywords and generate SEO-optimized listings.
- Built NestJS microservices with PostgreSQL and a React dashboard for listing review and publishing; maintained CI/CD pipelines.
- **Tech Stack:** NestJS, React, TypeScript, PostgreSQL, OpenAI API, Amazon Seller API

Software Engineer (iOS/macOS) — Jul 2024 – Jul 2025

KATco – Faisalabad, Pakistan

- Promoted from Associate Software Engineer. Built **PDF Converter** (App Store, iOS/macOS) end to end: image-to-PDF conversion, PDF rotation, and image resize, crop and watermarking; live in production with a 5.0 App Store rating.
- Cut manual processing effort by 40% via Cloud Converter API integration and improved processing speed by 30% with Swift Concurrency.
- Shipped **BGZoom**, a macOS video-calling app with real-time background effects for Zoom, Meet, Teams and Skype.
- **Tech Stack:** Swift, SwiftUI, UIKit, AppKit, Swift Concurrency

RESEARCH

Information Flow Guided Grey-Box Fuzzing — Aug 2023 – Apr 2024

Research Collaboration – DFKI Germany & FAST-NUCES

- Designed an algorithm combining static analysis with grey-box fuzzing to improve smart contract vulnerability detection while cutting resource usage by 25%.
- Validated results with researchers from DFKI Germany and FAST-NUCES and presented findings at both institutions.
- **Tech Stack:** Python, Solidity, Echidna, Foundry, Slither

EDUCATION

FAST-NUCES – BS Computer Science — Sep 2020 – Jun 2024

Faisalabad, Pakistan

- Dean's List (2x). Teaching Assistant for Artificial Intelligence and Physics; AI Programming Mentor at WEEPRO CODERS.